WHEN DOES CELL CONDITIONING HARM RECONSTRUCTION? DIAGNOSTICS FOR ARBITRARY-SCALE SUPER-RESOLUTION

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ABSTRACT

Cell conditioning in arbitrary-scale super-resolution can improve average quality while harming individual images or frequency bands. We connect failure diagnosis to a controlled test of how feature corrections are allocated. A frozen-cohort replication on 32 natural images, LTE and an LTE-derived experimental variant (SC-INR), under two inherited training seeds finds harm against zero and nonzero cell references. An exact squared-error decomposition separates overshoot from wrong-direction corrections, which positive output attenuation cannot turn beneficial. On a fixed $\times 4$ synthetic grid, we trace 16 SC wrong-direction cases. Restoring scale weights from a paired input rescues only one. In a separately frozen follow-up, replacing selective attenuation by uniform feature attenuation at exactly the native total feature-step norm restores positive middle-band gain in all 16 cases. This persists with zero-cell decoder ReLU gates frozen. Across all 32 middle-frequency conditions, the intervention makes every gain positive, including the 31 originally harmful cases. Yet it harms all 16 low-frequency controls in centre-region RGB. These results locate a content-dependent allocation failure in the specified model and probes, rather than establish a universal repair rule. Full-image RGB improvement in the focal cases is small: a median 0.40% MSE reduction over native conditioning.

***Index Terms*—** Arbitrary-scale super-resolution, cell conditioning, feature allocation, controlled failure diagnosis

1. INTRODUCTION AND RELATED WORK

Arbitrary-scale super-resolution (ASISR) predicts image values at continuous coordinates. LIIF supplies target cell size to a local decoder [1]; LTE introduces cell-dependent phase estimation [2]. These inputs specify the requested scale but do not guarantee that the resulting correction improves reconstruction. We ask: *can harmful cell corrections be explained only by their magnitude or decoder gate changes, or does allocation across latent features matter?* We first distinguish failure types, then intervene on the feature allocation of a specified failure class.

Harmful cell effects are not new. LIIF varies cell at fixed queries; IPE-LIIF integrates positional features and reports cases where adding cell hurts [3]. Thera supplies analytic Gaussian filtering and discusses mismatch with bicubic training [4]. Integrating latent features need not equal integrating a nonlinear decoder's RGB output. We diagnose inherited checkpoints, rather than benchmark performance against these methods.

Frequency-resolved analysis includes HyRA's linear and nonlinear responses [5], texture error analysis [6], sinusoidal probes [7], and Nyquist-normalised forecasting [8]. Phase beyond spectral magnitude is classical [9]. We claim neither these principles nor the quadratic identity below as new. Unlike architecture/retraining benchmarks [10], we test fixed-parameter interventions.

Our contribution is a bounded empirical chain: paired natural-image measurements establish failure types and their scale; controlled inputs identify a repeatable wrong-direction class; and a native-norm-matched feature intervention restores its gain even with decoder gates frozen. A low-frequency counterexample shows that the same intervention cannot universally replace selective conditioning.

2. MODELS AND FAILURE DIAGNOSIS

2.1. Fixed comparison and model lineage

Changing cell while holding LR observation, query and HR target fixed is a conditioning intervention, not a physical change of pixel footprint. Native cell does not certify an exact pixel integral. LTE and SC inherit an EDSR-baseline encoder, DIV2K training [11], a 256-dimensional representation, three 256-unit ReLU hidden layers and 1000 epochs at $\times 1$–$\times 4$. SC-INR is our earlier LTE-derived experimental variant, not an independent model family or a new method proposed here. LTE conditions Fourier phase on cell. SC predicts frequency ω_k and phase from image features, then weights sine/cosine features by

$$W_k(c_{\mathrm{rel}}) = \mathrm{sinc}(\omega_{kx} c_{\mathrm{rel},x}/2)\,\mathrm{sinc}(\omega_{ky} c_{\mathrm{rel},y}/2), \quad (1)$$

where $\mathrm{sinc}(v) = \sin(\pi v)/(\pi v)$ and $c_{\mathrm{rel}} = c\, n_{\mathrm{LR}}$. Its 128 frequencies use signed $2.1\tanh$ parameterisation. Cell enters only through W before the MLP, not the feature-predicted phase. Frequency/phase parameterisations also differ between LTE and SC, so cross-model quality cannot isolate sinc's effect.

A retrained no-cell ablation changes weights; our interventions do not. The natural-image replication uses inherited seeds 1 and 2; controlled probes use seed1. LTE snapshots omit the original extrapolation cell clamp. Zero cell is outside training support. Nonzero references test whether harm is confined to that contrast.

2.2. Overshoot, direction and the object being changed

For native prediction F_c, reference F_r, HR target t and $d = F_c - F_r$, set $a = \langle t - F_r, d\rangle$ and $b = \|d\|^2$. Inner products are pixel/channel means, optionally after band projection.

$$G(\lambda) = \mathrm{MSE}(F_r, t) - \mathrm{MSE}(F_r + \lambda d, t) = 2\lambda a - \lambda^2 b. \quad (2)$$

For harmful $G(1) < 0$, $0 < a < b/2$ is *overshoot*: a smaller positive correction can help. If $a < 0$, the *direction is wrong*: every positive multiple hurts. Returning to $\lambda = 0$ removes harm against the reference but yields no positive gain. This classical, HR-assisted diagnosis is not an observation-only method. Harm tolerance is 10^{-8}. Natural-output numerical boundaries use $64\epsilon_{64}\max(\mathrm{MSE}(F_r, t), b, 10^{-8})$ for b, $|a|$ and $|a - b/2|$; all boundaries are retained. Uniform *output* attenuation scales a fixed d; uniform *feature* attenuation below changes its decoded direction.

3. NATURAL-IMAGE FAILURES AND DAMAGE MAGNITUDE

We freeze DIV2K valid0837–0868 before inference: 32 images, LTE/SC seeds 1 and 2, and $\times 2/4$, excluding earlier development/forecast images. Centre HR128 targets and bicubic-AA observations are fixed. Relative cells are $\{0, 1/2, 2/3, 1\}$; native is 1 at $\times 2$ and $1/2$ at $\times 4$. All nonzero values occur numerically in training (LR48, integer HR48–192), but changing cell at a fixed target remains mismatched conditioning, not a joint in-distribution sample. Primary contrasts are native versus zero and versus $2/3$ ($h3$).

All 1024 predictions are hash-frozen before scoring. Primary metrics use unclipped RGB and shave4. Three non-DC Chebyshev bands have cutoffs $1/8, 1/4, 1/2$ and pixel-domain normalisation. Intervals use 10,000 image-cluster bootstrap resamples jointly across seeds; repeated seeds and scales are not independent images.

Benefit, harm and reference dependence. Against zero, mean PSNR improves by $.912/.444$ dB for LTE at $\times 2/4$ and $.687/.028$ dB for SC; all four 95% bootstrap intervals are positive. Table 1 nevertheless finds harm against both references and in both seeds. At $\times 4$, SC's eleven zero-reference RGB failures and 25 $h3$ failures are disjoint: failure belongs to a specified correction, not an intrinsic image label. Absolute quality is separate: native LTE/SC PSNR is 34.080/34.054 dB at $\times 2$ and 29.033/29.045 dB at $\times 4$; the latter paired difference has a 95% interval crossing zero. We claim no SC advantage.

Table 1. Natural-image replication. Each row has 64 RGB and 192 band records. RGB: negative records, then losses $> .01$ dB in parentheses. O/D: band overshoot/wrong-direction counts. $\Delta E_{1/2}$: mean RGB MSE change from native after halving the output correction.

Model	s	Ref.	RGB	Band O/D	$\Delta E_{1/2}$
LTE	2	zero	1 (1)	5/1	+7.55%
	2	$h3$	13 (6)	13/11	+2.29%
	4	zero	2 (1)	18/2	+2.77%
	4	$h3$	35 (10)	21/84	−.058%
SC	2	zero	10 (5)	10/12	+5.46%
	2	$h3$	17 (9)	13/16	+1.84%
	4	zero	11 (5)	17/30	+.264%
	4	$h3$	25 (7)	11/83	+.087%

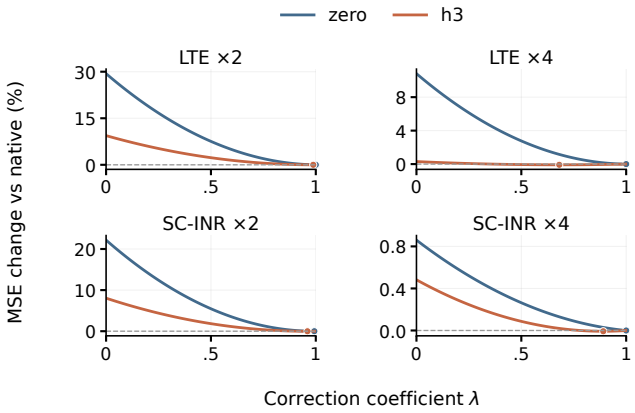


Fig. 1. Complete output-attenuation curves over 32 images and both seeds. Zero is native mean MSE; lower is better. Dots are exact HR-informed group optima, not deployable selections. Vertical ranges differ.

Attenuation has a small aggregate opportunity here. Retrospective complete curves use $\lambda = 0, .01, \ldots, 1$. The exact HR-informed uniform optimum is $\hat{\lambda} = \mathrm{clip}(A/B, 0, 1)$, with $A = \mathrm{mean}(a)$ and $B = \mathrm{mean}(b)$. Its largest mean RGB MSE reduction from native is .0011% against zero and .0853% against $h3$; per-record oracles yield at most .072% and .298%. These are empirical net improvements in this interpolation family, not evidence that individual negative cases are unrepairable. Setting $\lambda = 0$ can always return them to the reference; reducing corrections also sacrifices existing benefits.

Robustness and visible scale. Six fixed conventions cross raw or clipped endpoints with shave0/4/8; clipping precedes interpolation. All retain mean benefit over zero and overall harm from half-correction (MSE increases .226–7.67% across groups/conventions), although individual labels vary. Most negative RGB records in Table 1 lose at most .01 dB; counts do not imply large perceptual damage.

4. CONTROLLED FAILURE AND ALLOCATION INTERVENTIONS

4.1. A fixed class of wrong-direction corrections

At $\times 4$, a monochrome HR64 target repeated across RGB is $t = .5 + L + H$. Low modes are $L = \{(2,0),(0,2),(2,2)\}$ and the middle templates are $\{(10,0),(12,0),(14,0)\}$ and $\{(10,0),(0,12),(10,10)\}$, in cycles per 64 pixels. Each cosine amplitude is .04. Periodic separable bicubic-AA A gives

$$y_\eta = A(.5 + L) + \eta A(H), \qquad \eta \in \{0, .5, 1\}. \tag{3}$$

Four high-branch phases (zero plus three uniform vectors, seed 20261012) cross four translations $\{0, .5\}^2$ in LR-pixel units; both branches translate. High templates use $\{(18,0),(22,0),(26,0)\}$ and $\{(18,0),(0,22),(18,18)\}$. The low control uses $\{(4,0),(0,4),(4,4)\}$ and always observes At. The original 240 conditions contain 148 unique inputs, not independent images. Primary centre32 and sensitivity full64 contain integer periods; metrics are unclipped, with the same three bands.

For the 32 middle conditions, $\eta : 0 \to 1$ lowers middle-band gain in 29/32 LTE and 32/32 SC cases; alignment changes positive-to-negative in 14/32 and 16/32. The latter 16 define the focal SC class below; all 32 remain in every analysis. Direction/gain tolerances are $10^{-10}/10^{-8}$. Changing η fixes the target but changes LR energy and gradients; it is not pure aliasing. Equal-target-power phase changes in high templates produce both gain signs in all eight template–position groups per model: power alone does not determine benefit. At $\eta = 0$, hidden-phase variants share the same input/output and cannot demonstrate neural adaptation to hidden phase.

4.2. Test weight drift, then isolate allocation

The following tests reuse the seen grid and SC seed1 checkpoint; they are retrospective interventions, not new held-out image tests. In all 16 focal cases, the old correction remains positively aligned against the new zero-cell residual, whereas the new correction is negatively aligned against both residuals. A change in the remaining error alone therefore does not explain these reversals. Let U_i be each input's unweighted Fourier/coefficient features at zero cell, W_i its native sinc weights, and D the fixed MLP. First cross carriers U_0/U_1 and envelopes W_0/W_1, keeping recipient phase, carrier frequency, skip and decoder fixed. Swapping W_0 onto U_1 rescues only 1/16 focal negative gains. This swap does not support envelope drift as a major explanation. Two controls matched to the $U_1(W_0 - 1)$ feature-step norm—uniform attenuation and a fixed paired sine/cosine channel permutation—restore positive gain in all 16. This donor-norm observation motivates a separately frozen native-norm confirmation; it does not establish it.

For each $\eta = 1$ input, set

$$\Delta z_n = U \odot (W_1 - 1), \quad \Delta z_u = -\alpha U, \quad \alpha = \frac{\|\Delta z_n\|_F}{\|U\|_F}. \tag{4}$$

The norm spans four local-ensemble branches, 4096 queries and 256 features: $\|\Delta z_u\|_F = \|\Delta z_n\|_F$. It does not match each position/channel separately or match RGB correction magnitude. Both preserve U, skip, ensemble weights and MLP parameters. Their full RGB changes, with branch weights q_j, are

$$d_v = \tfrac{1}{2} \sum_j q_j [D(U_j + \Delta z_{v,j}) - D(U_j)], \quad v \in \{n, u\}. \tag{5}$$

The factor $1/2$ undoes output normalisation; both share the zero-cell RGB reference. A second response freezes all ReLU masks at U_j: replace the bracket by $J_D(U_j)\Delta z_{v,j}$. This is an exact fixed-mask affine intervention, not an approximation asserted for the original nonlinear network. It tests whether gate switching is necessary for the allocation effect.

Allocation changes the outcome at equal feature-step norm. In the focal 16 cases, native and frozen-mask native responses both have negative direction and gain. Uniform allocation makes both positive in 16/16 (Fig. 2). Over all 32 middle conditions, both decoder responses change from 31 negative gains to 32 positive gains. Relative to native, full64 RGB MSE improves in every focal case, with median reduction .3996% ($+.01739$ dB); centre32 RGB medians are .3724% and $+.01620$ dB. Restoration is not confined to the selected band, but its absolute performance effect remains small.

A content-dependent effect, not a replacement algorithm. All 16 low controls have positive native centre-RGB gain and negative uniform gain under both decoder responses. In 11/16 uniform cases the direction remains positive: overshoot accounts for most control failures. The intervention isolates allocation across feature/query positions as an intervenable factor in the specified class, beyond total feature-step norm and gate switching alone. It does not identify useful individual channels, establish learned frequencies as true image frequencies, or show that replacing selective conditioning is generally better.

5. SCOPE AND CONCLUSION

Natural-image replication supports failure measurements for the declared LTE lineage and contrasts; the allocation intervention covers one SC checkpoint, $\times 4$ and known synthetic samples. Zero cell is outside training support. Neither establishes natural-image causal prevalence or an observation-only selector. HR-assisted decisions are not unbiased risk estimation under generalized SURE assumptions [12]. No matching training was used. Each allocation stage freezes predictions before scoring; independent NumPy audits re-score every pixel

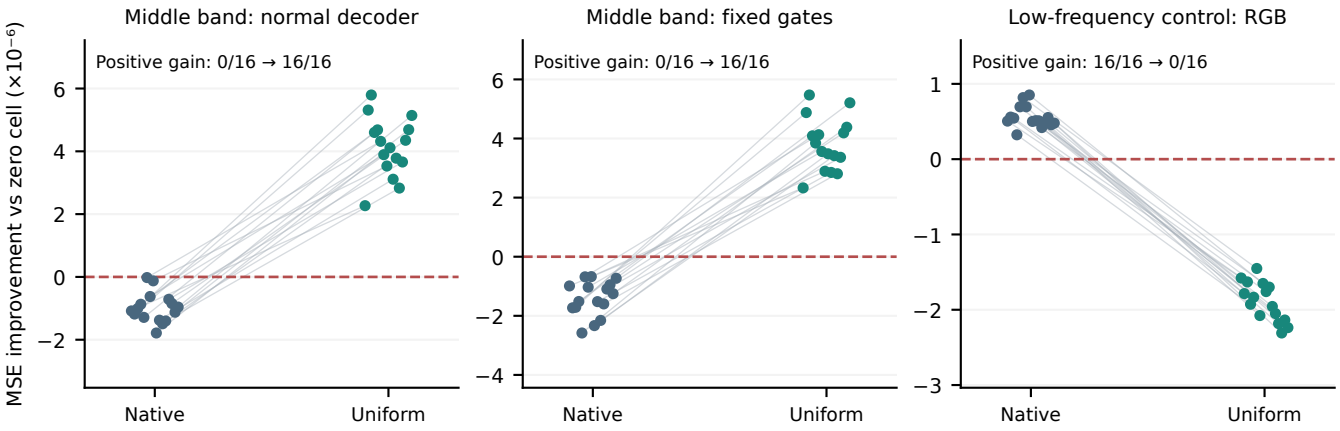


Fig. 2. Same native feature-step norm, different allocation. Each line is one fixed condition. Left: all 16 focal middle-band corrections switch from harmful to beneficial. Centre: this persists with zero-cell decoder gates frozen. Right: all 16 low-frequency controls instead switch from beneficial to harmful in RGB. All panels use centre32; the first two score the middle band, the third all RGB. Gain is relative to zero cell; vertical ranges differ. This is not a general repair.

and verify saved 16-query branch probes, not every internal feature tensor. The stages take 119.49 and 64.14 seconds on one CPU thread, without GPU.

Native conditioning can improve averages while individual corrections overshoot or move away from the target. The controlled SC intervention goes beyond coexistence: reallocating an equal total feature step restores a specified failure class, even without ReLU gate changes. The low-frequency counterexample makes the boundary substantive: appropriate allocation depends on content and the existing decoder. The contribution is this bounded diagnosis and intervention evidence, not a universally effective reconstruction method.

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7. REFERENCES

[1] Yinbo Chen, Sifei Liu, and Xiaolong Wang, "Learning continuous image representation with local implicit image function," in *CVPR*, 2021.

[2] Jaewon Lee and Kyong Hwan Jin, "Local texture estimator for implicit representation function," in *CVPR*, 2022.

[3] Ying-Tian Liu, Yuan-Chen Guo, and Song-Hai Zhang, "Enhancing multi-scale implicit learning in image super-resolution with integrated positional encoding," arXiv:2112.05756, 2021.

[4] Alexander Becker, Rodrigo Caye Daudt, Dominik Narnhofer, Torben Peters, Nando Metzger, Jan Dirk Wegner, and Konrad Schindler, "Thera: Aliasing-free arbitrary-scale super-resolution with neural heat fields," *Transactions on Machine Learning Research*, 2025.

[5] Haoyu Deng, Zijing Xu, Yule Duan, Xiao Wu, Wenjie Shu, and Liang-Jian Deng, "Exploring the low-pass filtering behavior in image super-resolution," in *ICML*, 2024.

[6] Salma Abdel Magid, Zudi Lin, Donglai Wei, Yulun Zhang, Jinjin Gu, and Hanspeter Pfister, "Texture-based error analysis for image super-resolution," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 2118–2127.

[7] Shusei Kawai, Fumiya Arai, Takao Marukame, Tetsuya Asai, Alexandre Schmid, and Kota Ando, "Frequency-response analysis of deep super-resolution models and visualization of degradation mismatch," *Nonlinear Theory and Its Applications, IEICE*, vol. 17, no. 4, pp. 1418–1433, 2026.

[8] Wenshuo Wang and Fan Zhang, "Breaking scale anchoring: Frequency representation learning for accurate high-resolution inference from low-resolution training," arXiv:2512.05132v2, 2026.

[9] Alan V. Oppenheim and Jae S. Lim, "The importance of phase in signals," *Proceedings of the IEEE*, vol. 69, no. 5, pp. 529–541, 1981.

[10] Tayyab Nasir, Daochang Liu, and Ajmal Mian, "Implicit neural representation-based continuous single image super-resolution: An empirical benchmark," arXiv:2601.17723v3, 2026.

[11] Eirikur Agustsson and Radu Timofte, "Ntire 2017 challenge on single image super-resolution: Dataset and study," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017, pp. 126–135.

[12] Yonina C. Eldar, "Generalized sure for exponential families: Applications to regularization," arXiv:0804.3010v2, 2008.